

Townsend 7.3-7.7

questions for gallicchio

7.3

$$\hat{a}^\dagger \rightarrow \begin{pmatrix} 0 & 0 & 0 & \dots \\ \sqrt{1} & 0 & 0 & \\ 0 & \sqrt{2} & 0 & \\ 0 & 0 & \sqrt{3} & \\ \vdots & & & \end{pmatrix}$$

$$\hat{a} \rightarrow \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \dots \\ 0 & 0 & \sqrt{2} & 0 & \\ 0 & 0 & 0 & \sqrt{3} & \\ \vdots & & & & \end{pmatrix}$$

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$$

7.4

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p}_x \right)$$

Once we have determined the ground-state wave function, we can take advantage of (7.37) to determine all of the position-space energy eigenfunctions:

$$\begin{aligned} \langle x|n\rangle &= \frac{1}{\sqrt{n!}} \langle x|(\hat{a}^\dagger)^n|0\rangle \\ &= \frac{1}{\sqrt{n!}} \left(\sqrt{\frac{m\omega}{2\hbar}} \right)^n \left(x - \frac{\hbar}{m\omega} \frac{d}{dx} \right)^n \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-m\omega x^2/2\hbar} \end{aligned} \quad (7.45)$$

For example,

$$\langle x|1\rangle = \left[\frac{4}{\pi} \left(\frac{m\omega}{\hbar} \right)^3 \right]^{1/4} x e^{-m\omega x^2/2\hbar} \quad (7.46)$$

$$\langle x|2\rangle = \left(\frac{m\omega}{4\pi\hbar} \right)^{1/4} \left(2 \frac{m\omega}{\hbar} x^2 - 1 \right) e^{-m\omega x^2/2\hbar} \quad (7.47)$$

7.5

Uncertainty relations tell us about the states of matter accessible - higher temperature -> higher energy -> higher uncertainty -> if on the order of internuclear distance, gas state. Everything is in a superposition, but interaction depends on the uncertainty / standard deviation in position. Compressing lowers uncertainty in x and increases the momentum, so things collapse to high

energy solids???

A good illustration of the effects of this zero-point energy is the unusual behavior of helium. Helium is the only substance that does not solidify at sufficiently low temperatures at atmospheric pressure. Rather, it is necessary to apply a pressure of

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at least 25 atmospheres. For substances other than helium, the uncertainty in the position of the nuclei in the ground state is in general quite small compared to the spacing between the nuclei, which is why these substances solidify at atmospheric pressure at sufficiently low temperature. In fact, increasing the temperature populates the higher vibrational states and increases the uncertainty, as (7.56) indicates. These substances melt when the uncertainty becomes comparable to the spacing between the nuclei in the solid. For helium, even in the ground state, the uncertainty is large because of two factors: the small mass of helium and the small value of ω (because of weak attraction between the helium atoms in the solid). Thus Δx is too large for helium to solidify at low temperature at a pressure of one atmosphere. Increasing the pressure reduces the separation between the helium atoms, thereby increasing ω and reducing Δx , so that at high pressure helium solidifies.

7.6

superpositions time evolve